

Steering Wheel

Solar Gators Steering Wheel

As part of the Cockpit subsystem at Solar Gators, I'm redesigning the steering wheel for our solar powered car. Taking F-1 steering wheels and our past designs as inspiration, a design was first drafted making sure it followed the Formula Sun Grand Prix (FSGP) regulations. The steering wheel for our next car will include a small screen as well as 8 actionable buttons and paddle shifters to activate regenerative braking.



Fig. 1. F-1 steering Wheel to the left, 2024-25 Flare steering wheel to the right.

The main changes the new steering wheel would have are the implementation of the screen on the wheel and paddle levers. Separate holes for the thumb will also be implemented to make steering and actuating the levers easier. The quick release will stay at the center of the wheel.

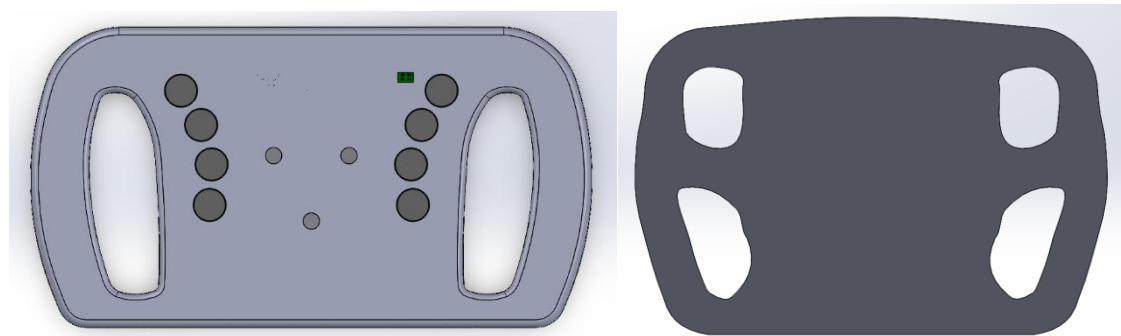


Fig. 2. 24-25 Flare Steering wheel design to the left, 25-26 preliminary design to the right.

The PCB that will handle the electric components will also be redesigned for this new steering wheel. The button placement will be chosen so it's reachable with the new design and a slot for the screen will also be added.

A prototype was printed to test how comfortable the shape of the new design was before adding the button locations and moving forward with the design in general.



Fig. 3. 3D Printed prototype.

After evaluating the prototype, it was determined that the screen should be moved up so that the quick release is in the center of the steering wheel. The size is optimal and fits in the dimensions of the new car and it feels comfortable when grabbing it. The next steps are to make the changes made above and design the button placement.

The paddle shifters will be attached at the back of the steering wheel and will be used for regenerative braking. This new addition has never been made before in our club, so more iterations are expected than usual.

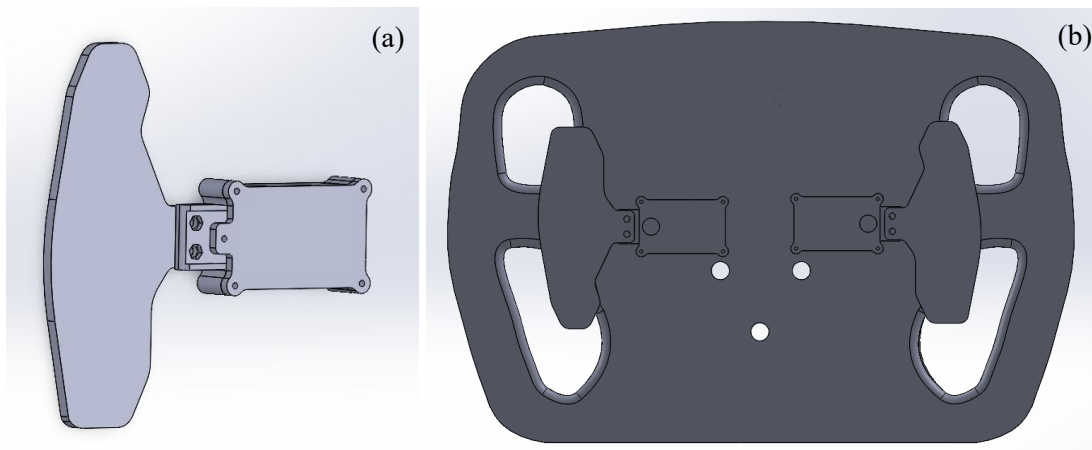


Fig. 4. Paddle shifter (a). Steering wheel assembly with preliminary paddle shifter positions (b).

After a mechanical design review, me and my team concluded that the quick release holes should be moved to the center of the wheel to improve the ease of rotation. Additionally, The screen was moved closer to the top edge to account for the quick release change and to make the design for PCB simpler. I also talked to our designated drivers and most of them didn't like how the slight bump on the handles felt. The arc was reduced significantly to make it as comfortable as possible for the users. Additionally, the drivers also thought the size was slightly too big, so material was cut from the bottom of the the steering wheel shape shown in Fig. 3. The result of the above mentioned iterations be seen below.



Fig. 5. Paddle shifter prototype (a). Steering Wheel model with modifications (b).

A prototype for the quick release was 3D printed and tested. The result was a working shifter composed of an arm hinged at one side with magnets on the arm and on the casing. When the arm was pressed, it pressed onto the tilt switch activating it. Then the magnets would attract each other, making the arm return to its resting position.

Once the general shape of the steering wheel was determined, the manufacturing process had to be determined. Previously, the team has used single plate and double plate carbon fiber wheels. For the single plate design, a carbon fiber sheet was bought and water jetted into the desired shape, and a casing was 3D printed to cover the wiring and the PCB for both safety and aesthetic reasons. For the double plate, a carbon fiber wet layup procedure was followed, using 3D printed spacers as insulation between the carbon fiber plates and the PCB, since carbon fiber is a conductor.

It was decided that the new design would follow the double plate concept, so new parameters were taken into consideration. Firstly, the LCD screen would take up space, and the current back plate design didn't account for that. Additionally, the button holes needed to be added and the fillets shown in the above designs were removed since it made manufacturing a longer process, and didn't provide any mechanical or design benefits. Hand supports were designed to optimize the ergonomics of the steering wheel and reduce driver fatigue.

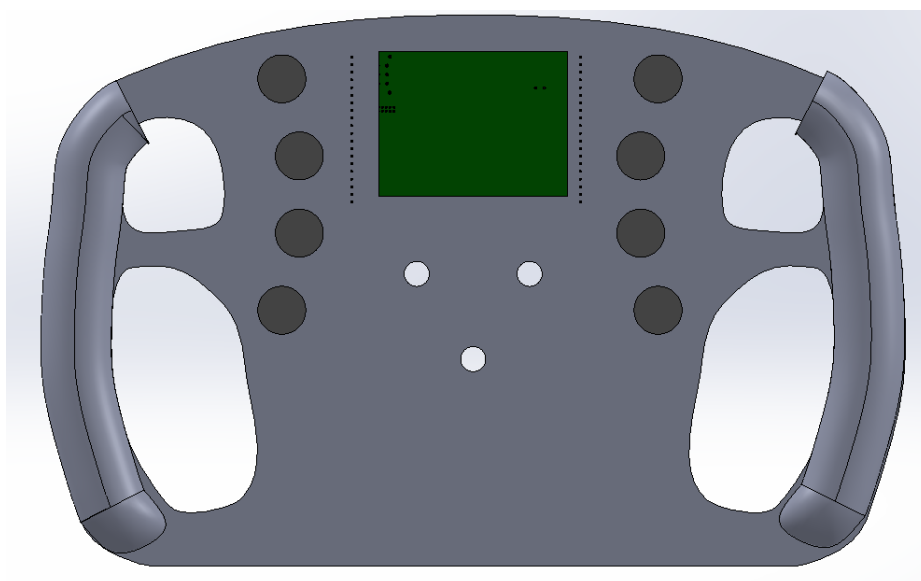


Fig. 6. Updated design with button placements and preliminary hand support designs.

